Muhammad Hamza Tahir

Development Engineer · Firmware, Mobile and Data

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SUMMARY

I build products end to end, from ESP32 firmware to the mobile app that talks to it. At Robusys I wrote the device firmware and I'm the sole mobile engineer on its companion app, which I took from an empty project to release on Android and iOS. IBM-certified Data Analyst who built an Airflow-orchestrated fraud detection pipeline in Python and PostgreSQL.

SKILLS

Languages	C, C#, Python, SQL, JavaScript
Embedded	ESP32, FreeRTOS, Wi-Fi, BLE, packet-based protocols
Mobile	.NET MAUI, XAML, MVVM, iOS (AccessorySetupKit, NEHotspotConfigurationManager), Android, biometric auth
Data	pandas, scikit-learn, PostgreSQL, ETL, Airflow, Jupyter, Excel, Cognos, Plotly, Streamlit
Web & tools	Django, DRF, REST APIs, WebSockets, React, Three.js, Docker, Git

EXPERIENCE

Development Engineer, Robusys · Lahore (on-site) — Sep 2025 to Present

Device firmware (ESP32)

- Wrote the ESP32 firmware from an early prototype, including a packet-based command protocol for communication with the mobile app.
- Built the command parsing and execution layer, tuned memory use and message flow, and ship features and field fixes each release.

Companion mobile app (.NET MAUI, C#, XAML)

- Sole mobile engineer: took a cross-platform app from an empty project to release on Android 10+ and iOS 18+.
- Designed an MVVM architecture with interface-based services and platform-specific code; built custom controls where stock MAUI fell short.
- Added Admin and Standard User roles with role-scoped biometric login and per-user credentials.
- Reached full iOS parity after Android launch: UINavigationController stack sync, a custom swipe-back gesture, provisioning profiles, and device pairing via AccessorySetupKit and NEHotspotConfigurationManager.

Patent documentation

- Drafted technical documentation and claims for the company's patent filings with the engineering team.

Technical Research and Development Associate, Logithoroughfare AB · Sweden (remote) — Jan 2024 to Aug 2025

- Researched technical and market questions using 20+ sources; wrote technical documents, process flows and first patent drafts; built landing pages and internal tools.

PROJECTS

Welfare Fraud Detection Pipeline `Python · PostgreSQL · Airflow · Docker · scikit-learn · Streamlit` — 2026

- 10-stage pipeline joining three disconnected sources (5,090 beneficiary, 18,844 payment, 6,040 national ID records); pandas ETL that flags instead of deleting, loaded into PostgreSQL.
- Benchmarked rules, RapidFuzz record linkage and Isolation Forest against 390 planted cases: 99% ensemble recall, 40/40 duplicate identities, Isolation Forest recall 12% → 84% (ROC-AUC 0.972).
- Orchestrated as an Airflow DAG in Docker with a regression quality gate and a Streamlit dashboard.

Smart Solar Inverter, real-time IoT monitoring `ESP32 · Django` — 2024 to 2025

- AC/DC power monitoring, from ESP32 firmware to a live Django dashboard with under 2 s latency.

ExplorOut, freelance e-commerce platform `Django` · **Zeroth**, 3D portfolio `React · Three.js · DRF` — 2025

CERTIFICATION

IBM Data Analyst Professional Certificate, IBM via Coursera · Aug 2026
3-month training by EXD under NAVTTC · credential BX5HAG96Y4E9

EDUCATION

BS Computer Science, The Islamia University of Bahawalpur · 2025
CGPA 3.32 / 4.00